

Securing Our Future: How the U.S. Can Foster and Retain Domestic STEM Students for
National Defense

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Abstract

The United States is currently lacking enough domestic students pursuing STEM degrees and careers, requiring the Department of Defense and its contractors to take proactive measures to increase engagement. This shortage threatens our nation's place on the global STEM stage and its security from foreign attacks, as only United States citizens can obtain the necessary clearances to work on defense projects. My thesis explains one of the main causes of the shortage, which is the lack of student awareness of potential STEM careers. The thesis then discusses the implications for the STEM workforce, specifically the defense industry, which is facing an employment crisis as a direct result of this shortage. This paper also reviews existing studies on increasing student participation in STEM and recommends two strategies for bolstering domestic STEM talent, both in universities and the workforce. First, it proposes that STEM companies should work more closely with universities, using funding from the Department of Defense to establish Field Specific Degree Programs (FSDPs). Through FSDPs, students will have more options for STEM degrees, and easier entry to the STEM workforce. Second, the thesis recommends that the Department of Defense establish Domestic Student Pre-Screening (DSPS), so that students can begin the process of receiving their necessary security clearances to work on national defense projects while still in college. This thesis then concludes with a section addressing the feasibility of the proposed mitigation strategies and anticipating possible counterarguments, as well as with a section on recommendations for further research.

Introduction: Determining the Problem

Current Shortage of Domestic STEM Talent

The United States is currently being surpassed by its biggest competitor, China, on the global stage for STEM. The Global Policy Institute noted in a 2024 article that “China is massively ahead of the U.S. in the number of schools, research centers, scientists and university graduates involved in the STEM disciplines,” (von Schirach *China Beats America in STEM Education*). This is due largely to the fact that “...a majority of Chinese students these days choose STEM university degrees, often supported by state of the arts laboratories and research facilities.” In America, however, “...only a small percentage of students prefer STEM disciplines,” (von Schirach *China Beats America in STEM Education*). Currently, the United States is suffering from a shortage of domestic students pursuing degrees in STEM. This has, in turn, led to a lack of qualified domestic STEM workers, especially in the national defense industry. The reason this is problematic is that American citizenship is a prerequisite of obtaining the security clearances necessary to work on national defense projects. As such, a shortage of domestic STEM workers will lead to significant delays and obstacles for the development of critical defense technology. Thus, it is of utmost importance that this issue is addressed. If this shortage continues, the United States will continue to fall behind China and other global competitors in STEM. Further, without a robust defense industry, our national security will become unstable, and the country will fall under risk of foreign attack.

Numerous studies have offered potential solutions to address this problematic shortage of domestic STEM students and workers. My work seeks to parallel other existing studies and provide a potential strategy to mitigate this shortage. In this paper, I put forth two strategies that the United States Department of Defense (DoD) could implement to increase the number of

domestic students pursuing STEM degrees and careers. First is funding STEM companies to partner with universities to establish Field Specific Degree Programs (FSDPs) and providing companies the resources to offer domestic employees government-level benefits. Second is to initiate Domestic Student Pre-Screening (DSPA). The implementation of these two strategies will accomplish three ends: help universities attract and retain domestic students; supply STEM companies, defense contractors, and the DoD with qualified domestic workers; and enable domestic students to obtain their necessary DoD security clearances sooner.

Section 1: Causes of Shortage

Student Awareness of STEM Careers

There are an array of factors causing the shortage of domestic students in STEM, though discussing all these factors falls outside the scope of this paper. As such, I will specifically address students' awareness of potential STEM careers, and how it contributes to the shortage, since this is the most pertinent to my thesis. One of the predominant factors in students' likelihood of pursuing a degree in a given field is their awareness of its associated careers. A study published in June of 2023 by researchers Emily Rosenzweig and Xiao-Yin Chen examined how high school students' awareness of the various types of jobs within STEM affected their desire to pursue a STEM degree or career. The results of the study showed that the majority of non-major STEM careers were unknown to students, noting that "...students have little exposure to a large number of specific STEM career options and do not consider them as readily as they do other more well-known STEM careers." Ironically, many of the overlooked STEM career paths were lucrative and in high demand (Rosenzweig and Chen *Which STEM Careers are Most Appealing?*). Rosenzweig and Chen's study shows that many students are unaware of the vast number of possible careers within STEM before they have even entered college. This puts students at a disadvantage when deciding whether to pursue a STEM degree.

This is especially the case considering there are numerous STEM fields with many career paths that are likely to be unknown to the majority of students. For instance, there are upwards of twenty-three different careers one can pursue with a computer science degree alone. Computer science is a vague degree to begin with, as one does not graduate and work as a "computer scientist." Rather, one would work as a software developer, systems administrator, data scientist, or even something as obscure as a cloud solutions architect (*23 Careers You Can Get with a*

Computer Science Degree).¹ Given this, let us consider two scenarios. The first is a high school student interested in pursuing a computer science degree but decides against doing so due to a lack of awareness of potential job opportunities within STEM. The second is a student interested in pursuing a computer science degree and follows through with it because of the vast potential employment opportunities. The Rosenzweig and Chen study shows that there are more instances of the former scenario than the latter.

Furthermore, regarding declining student interest in pursuing STEM careers, researchers Cohen et al. observed in a study that, “This decline appears to be due in part to a lack of awareness of the wide range of careers that students can enter with a science background. For example, 18% of students in a recent Harris Poll cited a lack of knowledge about health science careers as the reason for low interest (USciences, 2012),” (Cohen et al. *Fostering Stem Career Awareness*). In fact, research has shown that lack of career awareness is one of the main reasons students do not pursue a STEM career. A 2016 Emerson survey found that “...42% of adults would have considered a STEM career if they had better understood the opportunities available.” Moreover, “In the May 2022 report, Question the Quo, by the ECMC Group, a majority (63%) of teens said they wished they were provided more information about the variety of postsecondary opportunities available.” (Smith *STEM Career Awareness for Younger Students Enables Academic and Workforce Success*). Another recent study, *Voices of Gen Z: Perspectives on STEM Education and Careers*, found that merely 29 percent of students born between 1997 and 2012 list a STEM job as their most preferred career choice. This is despite the fact that 75 percent expressed an interest in areas of STEM. “The survey...found that lack of exposure to STEM subjects and opportunities...is partly to blame for the statistical disconnect, and that

¹ A cloud solutions architect works to ensure that a company is able to adopt and utilize cloud technology as smoothly as possible (23 *Careers You Can Get with a Computer Science Degree*)

students who are exposed to a higher number of STEM topics in middle and high school are more than twice as likely to pursue a STEM job or STEM major in college,” (Paykamian *Study: Gen Z Interested in STEM but Lacks Exposure*). The shortage of domestic STEM students resulting from insufficient career awareness poses major challenges for the STEM workforce, especially areas already suffering from the shortage of domestic students and workers, such as the defense industry.

Section 2: Assessing the Implications

How the Shortage of Domestic STEM Students Impacts the Workforce

The current lack of domestic students pursuing STEM degrees has numerous implications for certain STEM fields. For the purposes of this thesis, I will focus on perhaps the most critical – the impact on the national defense industry. The Stevens Institute of Technology noted in a 2023 article that “...research shows that fewer than 40% of students who begin a STEM program ultimately graduate with a degree in their chosen field, while about a fifth drop out of college altogether” (*Why don't students stick with STEM degrees?*). Comparatively, of the students who do pursue a STEM degree, even fewer graduate with one. According to data gathered by the Center for Security and Emerging Technology (CSET) in late 2023, “...just 20 percent of graduates in the United States obtain a STEM degree...” (Olis *The Global Distribution of STEM Graduates: Which Countries Lead the Way?*).

Another aspect of the issue is that the majority of STEM graduates in America are foreign students, particularly at the graduate level. Data collected from the National Foundation for American Policy (NFAP) show that “International students account for 70 percent of the full-time graduate students (master’s and Ph.D.s) in electrical engineering, 63 percent in computer science, 60 percent in industrial engineering, and more than 50 percent in economics, chemical engineering, materials engineering and mechanical engineering,” (*National Foundation for American Policy - NFAP policy brief*). This is not a problem in and of itself, as foreign students and workers certainly help bolster the STEM industry in many ways. However, according to these data, domestic students account for only between 30 to 50 percent of graduate students in STEM. When examined within both the broader global context, as well as the context of the defense industry, these data become problematic.

As of 2023, China and Russia were the world leaders in terms of the percentage of students obtaining STEM degrees (as a percentage of overall graduates). The United States was fourth from last. The data show that 41% of students in China and 37% in Russia obtain a STEM degree, versus only 20% of students in the United States. (Olis *The Global Distribution of STEM Graduates: Which Countries Lead the Way?*). Furthermore, “The population of China is four times that of the United States but is producing eight times the number of STEM graduates while Russia has almost four times the number of engineers,” according to a 2021 report released by the Pentagon regarding the state of the United States defense industry base (DIB), (Turner *Report: Pentagon says STEM education deficit is weakening America*). These statistics are deeply concerning, given the United States is graduating 21% less STEM students than China, and 17% less than Russia. The United States needs to rectify this discrepancy in the percentages of STEM graduates, as the lack of domestic students pursuing STEM degrees has led to a deficiency of qualified employees in the STEM fields.

Impact on Defense Industry

Domestic students’ inadequate participation in STEM leads to a shortage of qualified STEM workers, especially in the defense industry. While it is the case that the high percentage of foreign STEM students and workers positively impacts the STEM industry overall, the shortage of domestic STEM workers is problematic for the United States, particularly for the defense industry. The reasoning is that the United States Department of Defense (DoD) requires any individual working on defense projects to have specific security clearances, which can only be obtained by United States citizens. Thus, if too few domestic students are graduating with STEM degrees and entering the STEM workforce, certain areas of STEM, such as the defense industry,

will face an employment shortage. Even if one were to argue that this is not currently the case, it is clear that it will be in the near future.

In addition to the low number of domestic STEM graduates, almost half of the current domestic STEM workforce is beginning to age out and retire. According to 2021 data collected by the United States Census Bureau, 40.6% of the STEM workforce was comprised of employees ages 45 and over (Laughlin et al. *Technician Is a Common Job Among Young STEM Workers*). This means that in as few five to twenty years, a significant portion of the STEM workforce will need to be replaced. With few STEM graduates, the foreseeable gaps in the workforce will be difficult to fill. To reiterate, only 20% of domestic students are obtaining STEM degrees (Olis *The Global Distribution of STEM Graduates: Which Countries Lead the Way?*).

The Department of Defense has issued numerous publications on the current need for skilled STEM workers in the defense industry. In January of 2024, the DoD published a report in which they acknowledged the “...serious limitation in STEM ecosystems’ current ability to produce a STEM and skilled technical workforce for the defense industry...” (*Toward Collective Impact in DoD’s STEM Activities and Outreach*). Additionally, in 2023, the DoD published the “National Defense Industrial Strategy” (NDIS), a 59-page document aimed at ensuring the DIB will be able to meet the high demands of national security. The document highlighted that “...the U.S. labor market lacks sufficient workers with the right skills to meet domestic production and sustainment demand. For example, in 1985 the United States had 3 million workers in the defense industry. However, by 2021 the country had 1.1 million workers in the sector, a reduction of nearly two-thirds,” (Stewart *Vital Signs 2024: U.S. Defense Industrial Base Still Facing Headwinds*).

Furthermore, as of 2022, “82% of companies in the defense industrial base report that it is difficult to find qualified STEM workers,” according to a report from the Institute for Progress (IFP), a non-partisan think tank based out of Washington, D.C., (Neufeld *STEM Immigration Is Critical to American National Security*). Similarly, the United States National Science Foundation (NSF) found in their 2023 STEM workforce report that 74 percent of United States STEM workers do not possess a bachelor's degree. Similarly, data from a 2024 study conducted by NSF show that, of the 36,804,000 STEM workers in the United States, only 17,548,000 possessed a bachelor's degree or higher. Conversely, of the non-STEM workforce, 41,673,000 of the 115,564,000 workers possessed at least a bachelor's degree (Taylor and Arbeit *Science & Engineering Indicators*). While a higher percentage of STEM workers obtained at least a bachelor's degree than non-STEM workers in 2024, a much greater portion of the population was employed in non-STEM jobs versus STEM jobs. Especially considering the magnitude of the demand for qualified domestic STEM workers, it is clear the shortage of STEM graduates is a pressing and profound issue.

Dr. John Skrentny, a sociologist from Harvard, addressed this problem in his 2023 book *Wasted Education: How We Fail Our Grads in STEM*. In the introduction of his book, Skrentny pointed out that, “In 2022...US News said that by 2030, we would need 1 million more STEM grads, explaining, ‘Experts suggest there are not currently enough qualified STEM degree graduates to meet employment demands,’” (Skrentny 10). At the end of 2022, the United States Chamber of Commerce expressed a similar sentiment, noting, “There are currently more than 16 million skilled technical jobs requiring an associate degree or similar level qualification, and the number of jobs requiring substantial STEM expertise has grown nearly 34% over the past decade

(Boggs et al. *Addressing the Stem Workforce Shortage*). The Boston Consulting Group (BCG) also commented on this issue. BCG noted in a 2024 report that:

Every year, the U.S. will need about 400,000 new engineers. Yet the next-generation skill sets that those engineers will require are sorely lacking, presenting the alarming possibility that nearly one in three engineering roles will remain unfilled each year through at least 2030.” (Kodey et al. *The US Needs More Engineers. What's the Solution?*)

This lack of domestic STEM graduates equates to a shortage of qualified domestic workers in the defense industry. This is extremely problematic, as it is resulting in a deficiency of United States citizens qualified enough to work on critical national defense projects and technologies. The need for domestic workers in the defense industry is evident and imminent. If the current deficiency continues, the United States will be unable to keep up with leading competitors in terms of defense technology and capabilities, leaving the country vulnerable to foreign threats and invasion.

Section 3: A Review of Existing Literature

Proven Strategies for Increasing Citizen Participation and Proficiency in STEM

Given the significant consequences the domestic student shortage poses to the American defense industry and the United States' ability to compete on the global defense and STEM stages, developing strategies to improve domestic STEM participation is of paramount importance. Various researchers and organizations have already undertaken this task and offered significant contributions to existing literature. For instance, the Australian Council of Learned Academies (ACOLA), published a research study in 2013 that examined data gathered from twenty-two specific reports commissioned worldwide. The aim of each report was to see what methods leading STEM countries have and are implementing to foster participation and proficiency in the fields of STEM. The impetus for the study was to find strategies to improve STEM participation and proficiency in Australia, as the country is falling behind other developed nations, though their current state is less severe than that of the United States.

The study concluded several key findings, including numerous insights as to current STEM issues and challenges countries are faced with, and how STEM-strong countries, such as Finland, are tackling these challenges.² The methodology of the report involved commissioning consultants across twenty-two developed nations to report on a number of STEM related matters, including but not limited to: attitudes towards STEM; perceived relevance of STEM as important to economic development and sustainability; current patterns of STEM provision in schools; STEM graduates' access to labor markets; and strategies, policies, and programs to enhance all levels of STEM education (Marginson et al.). The findings of the report conclude that the most

² For reference, "STEM-strong" refers to a country with a robust STEM workforce and education system

common strategies across STEM-strong countries include high levels of investment in STEM education and pedagogy, fostering positive student attitudes towards and high awareness of STEM learning and careers, and building high-end research performance and doctoral student numbers in the fields of STEM (Marginson et al.).

Each of these common strategies can be further broken into their core tenants. High investment in STEM education can be divided into three main approaches: strong disciplinary frameworks; instilling problem solving, inquiry, critical thinking and creativity; and incorporating tasks into school science and mathematics curricula to foster students' problem-solving skills, flexibility in thinking, and awareness of engineering design activities. Investment in pedagogy can similarly be divided into two underlying approaches. The first is high regard for teachers, reinforced by meritocratic career structures built to recognize teaching excellence, and the second is high standards for teachers (Marginson et al.).

Fostering positive student attitudes towards STEM learning and high awareness of STEM careers was a significant contributor to STEM-strong countries' robust STEM education and workforce. As with investment in STEM education, there are three key strategies to encourage students to have positive attitudes towards STEM and to increase their awareness of the vast number of opportunities for careers in STEM. These include involving students' families in mathematics and science learning to build positive attitudes to careers in STEM; increasing opportunities for students and their families to positively interact with science and mathematics through events, exhibitions and other approaches; and programs which engage students in science or mathematics projects that relate to their local communities (Marginson et al.).

Finally, building high-end research performance and doctoral student numbers in the fields of STEM involves developing strategies and programs for forming, attracting, and

retaining highly skilled workers and researchers in the STEM fields. Countries seeking to strengthen their STEM sector often implement various strategies to attract talented STEM workers and researchers. One key strategy successful STEM countries have implemented to develop skilled STEM labor and research sectors is to form well-trained and talented STEM individuals through doctoral training programs. Leading STEM countries, such as China and Korea, all graduate high percentages of Science and Engineering (S&E) doctoral students each year (Marginson et al.). Thus, STEM-strong countries are able to ensure that their workers and researchers are highly trained, talented, and qualified to meet their nation's STEM needs.

Another report on developing a robust STEM education system is one published in October of 2022 by Yurdagül Boğar, a postdoctoral researcher, and Jari Lavonen, professor of Faculty of Educational Sciences at the University of Helsinki, Finland. Their report, entitled *Status and Trend of STEM Education in Finland*, thoroughly details the strategies Finland has and continues to implement to form and maintain their robust STEM education system and workforce. Finnish students have historically performed highly on the PISA administrations, frequently being ranked among the top five performers.³ Finland has taken many diverse approaches to improving and sustaining the quality of their STEM education.

Two of the strategies that Finland has implemented which stand out are their prioritization of including STEM-oriented curricula in schools and developing widespread interest and concern towards STEM among educators. An example of Finland's STEM-oriented curricula is their middle school curriculum, which is "...designed as an amalgam of three different curricula, biology, physics, and chemistry...Mathematics is introduced as a separate school subject and has links to several school subjects, not only STEM subjects. Engineering and

³ The Programme for International Student Assessment (PISA) is a standardized test administered to 37 countries worldwide, testing high school students on their reading, mathematics, and science comprehension and proficiency

technology are taught as a part of science subjects and as a part of craft subjects (Lavonen, 2021),” (Boğar and Lavonen *Status and Trends of STEM Education in Finland*). One of the benefits of STEM-oriented curricula is that it exposes students to STEM subjects at an early age, and continues to do so throughout their education, helping them to develop the skills they need to succeed in STEM subjects at the collegiate level and beyond.

Another benefit is that STEM-oriented curricula, such as is seen in Finland’s middle schools, is able to provide students the education they need to develop important STEM skills, but without detriment to other core subjects. For instance, the way in which Finland integrates STEM subjects with crucial mathematics and science subjects allows their educators to teach important math and science skills without detracting from the important STEM skills, and vice-versa. In regard to developing widespread interest and concern towards STEM among educators, Finland attempts to keep educators involved with STEM throughout their teaching career, and ensure they possess and maintain the STEM skills needed to effectively teach their students. Finland does so through various teacher programs, such as the LUMA Centre Finland. This program is aimed at improving “...both the lifelong learning and research-based teaching of the teachers working at different education levels,” as well as motivating “...students at different levels in science, mathematics, and technology learning,” (Boğar and Lavonen *Status and Trends of STEM Education in Finland*). The benefit of this is that it not only holds teachers to a certain standard but also provides them with the means of living up to that standard. Ultimately, the higher the quality of the teachers, the higher the quality of the student’s education.

The results of these two case-studies suggest that countries with robust STEM education systems and workforces invest highly in their schools and students at all educational levels. One of the most significant effects of their high investment in STEM education, especially at the

university level, is that it opens the way for students to enter the STEM workforce with minimal to no difficulty. Thus, such countries are able to develop and maintain strong STEM workforces that are well-supplied with capable and qualified workers. Based on the findings and conclusions of these two reports, namely the proven strategies STEM-strong countries implement to foster and sustain robust STEM education systems and workforces, I have developed several strategies to increase domestic student participation in STEM.

Section 4: Proposed Solution

Improving Domestic Student STEM Participation by Opening the STEM Pipeline

The severe lack of students who perceive themselves as capable of pursuing a degree or career in STEM is a serious and longstanding problem that will require various long term and likely expensive solutions. In the interim, it is possible for us to increase the percentage of capable students who do in fact pursue STEM degrees and careers. This can be done by increasing students' awareness of less well-known career paths within STEM and by making the paths to such careers simpler and better supported.

To mitigate the shortage of domestic STEM workers and students, I propose that the Department of Defense provides funding to STEM companies for the purpose of establishing better connections with universities. Specifically, STEM companies and universities should work in closer communication to determine the needs of the STEM workforce and how the universities can help. As part of this, STEM companies should use funding provided to them by the DoD to establish closer relations with universities. The purpose of this would be to help feed the STEM pipeline.⁴ As previously mentioned, the pipeline is currently supplying the STEM workforce with a high number of foreign workers, but not enough domestic workers. However, attracting more domestic students to STEM and establishing easier access for them to enter the workforce would help mitigate this problem. This is the aim of the strategies I put forth.

Similar partnerships to the kind I am proposing have been established in the past, and the benefits are evident. An instance of this is Sentient Digital and their subsidiary RDA partnering with Villanova University in Philadelphia, Pennsylvania. As noted by Sentient Digital, "The

⁴ The STEM "pipeline" refers to any source of STEM workers, such as universities

most obvious benefit [of partnerships] may be that universities can provide employers with a pool of highly skilled and motivated job candidates who are up to date with the latest knowledge in the field,” (Sentient Digital, Inc. *The Benefits of University Partnerships with Companies: Key Takeaways from Sentient Digital and RDA at Villanova*). The nature of company-university partnerships varies, though it often manifests in the form of research funding and student internships. While hundreds of such partnerships take place, and should certainly be encouraged, I am proposing a new type of partnership between STEM companies and universities, known as Field Specific Degree Programs (FSDPs). The University of Pennsylvania has done something similar to FSDPs with their systems engineering program. Due to their proximity to numerous engineering firms, the University of Pennsylvania established a systems engineering program to help supply these firms with capable students from their university. By establishing a program specifically for systems engineering, the University of Pennsylvania was able to offer their students a field specific degree that would grant them easier access to the STEM workforce through the nearby engineering firms. To more clearly explain the new type of partnership I am suggesting, I have broken FSDPs into three participating entities: companies and universities; domestic students; and the Department of Defense.

Companies and Universities

I believe one of the key ways to mitigate the lack of qualified domestic students entering the STEM workforce is for STEM companies to partner with universities to establish FSDPs. For instance, if Lockheed Martin were to partner with Carnegie Mellon University or the University of Pittsburgh to establish a defense engineering degree program, then Lockheed would have established a direct source of qualified domestic STEM workers. An FSDP partnership such as this would not only benefit companies by providing them with a direct source of qualified

employees, but it would also benefit the university students by providing them with easy access to the STEM workforce.

In addition to FSDPs, universities should work with STEM companies and the Department of Defense to develop certain certificate programs to better address the needs of the STEM workforce. Students can already obtain certificates in STEM fields such as engineering, cybersecurity, and data science. However, I suggest that universities communicate with STEM companies and the DoD to see which specific STEM fields require more workers or at least require more qualified workers. With this information, universities could then develop specific certificate programs for those fields. This would effectively accomplish the same end as FSDPs – developing a new and effective way to attract more domestic STEM students and workers.

Domestic Students

Another benefit of such a partnership is that Field Specific Degree Programs could encourage more domestic students to pursue STEM degrees. If a student can enter college with an effective guarantee of having a place in their desired STEM field, domestic students would be more likely to pursue and graduate with a STEM degree. This could also make STEM more attractive for students who are unsure as to whether they want to pursue a degree and career in STEM, as it would provide students with a more visible career path. If a student can see the type of job they can get with their field specific degree, the income of that position, the company they would work for, and so on, it would encourage them to follow through with obtaining their STEM degree and entering the STEM workforce.

Field Specific Degree Programs would also enable students to see the various types of jobs available to them within STEM. There is more to working a STEM job than a title that simply says “engineer,” or “software developer,” and so on. Yet many students are not aware of

the vast variety of options open to them within STEM. To reiterate, the Rosenzweig and Chen study, discussed in Section 1 of my paper, examined high school students' awareness of the variety of STEM careers, and which careers were most or least appealing. The study showed that students are unaware of the large number of STEM careers available to them even before they have entered college. This suggests that exposure to STEM at the high school level needs to be increased, but that matter falls outside the scope of this paper. With FSDPs, however, college students would be exposed to the lesser-known fields of STEM. This could increase domestic students' interest in pursuing STEM careers, as they would have a wider variety of degrees and careers to choose from, as opposed to only the more well-known STEM occupations.

Furthermore, the high wages for STEM jobs could also incentivize students to pursue a STEM career, especially if they were to have direct access to the workforce through FSDP partnerships. For reference, the median income for STEM occupations in the United States in 2024 was \$95,000, which is \$50,000 higher than the median income for non-STEM occupations (Grays *The STEM Landscape in 2024: Key Trends and Statistics*). Through FSDP partnerships, domestic students would be able to clearly see the wide variety of STEM degrees and opportunities available and pursue whichever one they desire. Then, upon graduation, students would assume a position in a STEM company making a significantly higher salary than if they were working a non-STEM occupation.

Department of Defense

One of the possible obstacles for companies to establish these partnerships with universities is funding, as such partnerships would likely be costly. This is where the Department of Defense comes in. Of all parties concerned, the DoD is most heavily affected by the current shortage of domestic STEM workers. While it's true that STEM companies are also affected, in

that they are lacking qualified employees, the DoD is lacking qualified contractors to ensure our nation's defense technology is ahead of our competitors. Should we fall too far behind our global competitors in this area, our nation will become less secure against foreign attacks. To clarify, security includes safety from foreign attacks of any kind, whether cyber, terrestrial, or aerial. As such, it is the DoD's responsibility to incentivize STEM companies to engage in these types of partnerships with universities, as well as incentivize domestic students to pursue STEM degrees and careers.

First and foremost, the Department should allocate a certain amount of funding to STEM companies to establish these partnerships. This way, STEM companies wouldn't have to draw from their own budgets to partner with universities. As shown in the studies discussed in my literature review section, STEM-strong countries invest highly in their STEM labor and research sectors. Providing STEM companies with funding would be one way the United States could imitate STEM-strong countries in this regard, as it would allow the government to invest in both the STEM workforce and education system simultaneously.

Beyond providing funding, the Department of Defense should also allocate STEM companies the resources to offer domestic workers similar benefits as government workers receive, which are substantial. Government employees often receive such benefits as student loan repayment and forgiveness, substantial health care insurance, and various comprehensive retirement plans (*Employee Benefits*). If STEM companies were able to offer domestic students similar benefits upon their graduation, this would incentivize more students to pursue STEM degrees and careers.

Another means the Department could employ to incentivize domestic students to pursue STEM careers would be to pre-screen them for security clearances. One of the barriers to

pursuing a STEM career, particularly in the defense industry, is the amount of time it takes to obtain the necessary security clearances from the DoD. While the most basic level of clearances (Interim/secret clearance) can take as short as one-two months, any level of clearance above Interim takes up to fifteen months (*Top Secret / Sensitive Compartmented Information (TS/SCI) Clearance*). The Center for Strategic and International Studies (CSIS) published a report in November of 2020 on how the DoD could attract and retain workers. In this report, the CSIS notes that “Compared to the private sector, where hiring is quick, open, and transparent, STEM professionals may view the defense enterprise’s onboarding process as a deterrent.” Consequently, STEM workers may leak out of the defense industry pipeline before even having a chance to fully enter it (Dwyer et al. *To Compete, Invest in People: Retaining the U.S. Defense Enterprise's Technical Workforce*).

The DoD could combat this potential leak by screening domestic STEM students before they enter the workforce, so as to accelerate the clearance process. This way, when a student graduates with a STEM degree, they are already eligible to receive their DoD security clearances and will be able to enter directly into the defense workforce. I suggest that to further streamline the process, Domestic Student Pre-Screening (DSPS), as I will refer to it, should be included in the FSDPs. At any point during their education, FSDP students who think they may want to work in the defense industry could apply for DSPS. If the student is approved for pre-screening, the DoD could begin the process of clearing the student to work on defense projects. Thus, FSDP students could obtain their security clearances with their diploma upon graduation and then enter directly into the STEM workforce. For instance, let’s say there’s a student attending the University of Pittsburgh, who is partnered with Lockheed Martin to offer an FSDP in defense engineering. The student could graduate with his degree in defense engineering, along with his

DoD security clearances, and begin working at Lockheed immediately out of college. Pre-screened students could even work for the DoD itself upon graduation. As such, Field Specific Degree Programs and Domestic Student Pre-Screening would incentivize domestic students to pursue both degrees and careers in STEM, supplying universities, STEM companies, and the Department of Defense with the capable students and workers they need.

Section 5: Addressing Feasibility and Potential Counterarguments

Key Questions and Concerns Regarding the Feasibility of the Proposed Solutions

There are several questions and potential problems that would need to first be addressed before the solutions I propose could be implemented. Most notable among these are security concerns with Domestic Student Pre-Screening and the long-term sustainability of Field Specific Degree Programs. Beginning with security of DSPS, there are concerns as to whether entrusting college graduates with clearances to access highly sensitive government information would pose a risk to national security. In recent years, there have been instances of classified government documents being leaked to the public by college students entrusted with this information.

As recently as April of 2023, a member of the United States Air National Guard (USANG), twenty-one-year-old Jack Teixeira, released government files regarding the war in Ukraine on the social media platform, Discord. The young college student was sentenced to fifteen years in prison for violation of the Espionage Act.⁵ Teixeira was an IT technician for the 102nd Intelligence Wing of USANG, a job which required him to be given Top Secret Information (TSI) clearance.

In court, Teixeira admitted to using his high-level clearance to unlawfully retain and transmit numerous classified documents (*Former Air National Guardsman Sentenced to 15 years in Prison for Unlawfully Disclosing Classified National Defense Information*). Such actions pose a risk to our nation's security, as our adversaries could obtain and utilize leaked classified

⁵ “The Espionage Act of 1917 prohibited obtaining information, recording pictures, or copying descriptions of any information relating to the national defense with intent or reason to believe that the information may be used for the injury of the United States or to the advantage of any foreign nation,” (Asp *Espionage act of 1917 (1917)*)

information. The concern with entrusting students upon their graduation with DoD security clearances is that they would abuse the privileges of these clearances, and similar instances as that with Teixeira would arise. The Department of Defense would need to find a way to ensure that college graduates would not abuse their clearances before implementing DSPS.

Another potential concern that would need to be addressed is the long-term sustainability of FSDPs⁶. In the event that the shortage of domestic STEM students is remedied, it may not be practical or sustainable for companies to continue engaging in FSDPs with universities. My response to this is that FSDPs do not need to be sustainable in the long term. The purpose of FSDPs is to help mitigate the current shortage of domestic STEM students. Should the issue of the shortage be resolved, FSDPs would no longer be necessary. This being said, if companies and universities would like to continue to implement FSDPs even after the shortage is no longer an issue, then long-term sustainability would largely be determined by the cost to both entities. It would likely be costly for companies and universities alike to implement FSDPs. If this is the case, then FSDPs would not be sustainable in the long-term, even with the DoD offsetting much of the cost to companies, who would, in turn, help defray the cost to universities. Although, to reiterate, long-term sustainability should not be concerning, as FSDPs are intended to only be a short-term implementation.

Responding to Potential Counterarguments

In addition to questions of feasibility, I will also anticipate and respond to potential counterarguments to various points made in this thesis. The first counterargument I will address is concerns that increasing domestic student participation in STEM will further feed what some

⁶ Merriam-Webster Dictionary defines long-term as “of, relating to, or constituting a financial operation or obligation based on a considerable term and especially one of more than 10 years.”

refer to as the “war machine,” that is, the United States military. My first response to this is that it is important to note the distinction between the national defense industry and the military.

While the defense industry does develop technologies and weaponry used by the military, it is still a completely separate entity from the military itself. This thesis does not once reference the military, advocate for increased militarization of the United States or an increase in military budget or enrollment or promote aggressive acts towards our global peers. Furthermore, there is yet another distinction between defending the nation and acting as a war machine. It is unlikely for a war or some other conflict to occur on United States soil, though not impossible. In such a case, the United States needs to be able to defend itself from attackers. Current events, such as the Russian invasion of Ukraine, demonstrate the importance of a nation’s capabilities to defend itself. Ukraine’s ability to defend against Russia, aided through foreign support, has proven crucial to maintaining its sovereignty. Though the United States should not act as an active military aggressor, it is also important for it to be capable of defending itself. To reiterate, then, this thesis does not advocate for feeding the “war machine,” but rather advocates for bolstering the United States’ national defense industry, for means of providing security to our nation against potential foreign attacks.

Another important counterargument I will refute is that the United States should simply rely on foreign workers to fill the gaps in the STEM workforce and defense industry, especially considering the high number of international students obtaining STEM degrees from American universities. My response to this is twofold. First, I will not deny the value of foreign workers to the STEM workforce. In many ways, they have helped sustain the United States’ STEM sector amidst domestic worker shortages, and the STEM industry would not be where it is without them. This brings me to my second point.

Despite the benefits foreign workers provide to certain areas of STEM, the United States cannot rely on them for the defense industry. No country would allow citizens from their global competitors to obtain highly classified information and work on critical national defense projects. China certainly would not allow a citizen from the United States to work on one of their defense projects. And neither should the United States. To disallow foreign individuals to work on national defense projects is not an act of xenophobia, but rather of security, practicality, and common sense.

This is not to say that all foreign workers should be treated as untrustworthy, but it is simply a risk the United States cannot afford to take. Granted, there are means by which foreign individuals can work for United States' DoD and its defense contractors, such as H-1B visas.⁷ However, these provide no level of security clearance, meaning H-1B holders cannot access classified information regarding national security, but can work in certain capacities on defense projects. It is also the case that Limited Access Authorization (LAA) "...may be granted in rare circumstances where a non-U.S. citizen possesses a unique or unusual skill, or has expertise that is urgently needed," (*Security Clearances for Non-U.S. Citizens*). Nevertheless, there is a difference between granting certain foreigners LAA in rare circumstances and permitting any foreign individual to work on classified national defense projects.

The final counterargument I will address is the fact that any of the United States' best domestic STEM workers are already entering defense engineering. While I could not find specific data supporting this claim, I will address this matter from two perspectives. First, if this is the case, then the defense industry is only the better off for it. Although it is clearly not enough

⁷ "The H-1B is a temporary (nonimmigrant) visa category that allows employers to petition for highly educated foreign professionals to work in "specialty occupations" that require at least a bachelor's degree or the equivalent" (*The H-1B visa program and its impact on the U.S. economy*)

to offset the shortage of domestic STEM workers in the defense industry. As such, the United States would need to bolster domestic employment in the defense industry regardless of whether or not its best STEM workers are already pursuing defense engineering. Second, if it is not the case, then the shortage of domestic STEM workers, particularly in the defense industry is only more pronounced, and the United States needs to urgently address it. Thus, from either perspective, the issue remains the same.

Section 6: Conclusion and Suggestions for Further Research

Conclusion

The United States is currently facing a shortage of domestic STEM students and workers. This shortage has several causes, though I focused on one that I found pertinent to this paper. The specific cause I examined was lack of student awareness of potential STEM careers. Students' unawareness of the vast number of STEM careers is leading to a lack of domestic students and workers qualified enough to meet the ever-increasing demands of the STEM workforce. One of the most severe implications of this shortage is that the defense industry is lacking qualified domestic engineers and other STEM employees to work on national defense projects. This has led to the United States falling behind global competitors, such as China, in the areas of STEM and national defense. Without enough qualified domestic workers in the defense industry, our national security falls under risk.

This paper also analyzed two research studies on proven strategies for increasing citizen participation in STEM. The first study analyzed twenty-two leading STEM countries and the strategies they implemented to develop and maintain robust STEM education systems and workforces. The second study analyzed Finland and the strategies it implemented to form its strong STEM sector. The results of these two studies showed that STEM-strong countries invest highly in their STEM workforces and education systems in a variety of ways.

In this thesis, I put forth two strategies of my own to help mitigate the shortage of domestic STEM students and workers. First, I propose that the Department of Defense should provide STEM companies with funding to establish Field Specific Degree Programs in universities, as well as the resources to offer government-level benefits to domestic employees.

Doing so would help attract and retain domestic STEM students. Second, the DoD should initiate Domestic Student Pre-Screening. Through DSPS, students can begin the process of obtaining their DoD security clearances sooner, reducing delays in the DoD onboarding process.

My contribution parallels similar works on this topic but is unique in that a solution like mine has yet to be suggested. Many solutions deal with increasing diversity, providing more funding to high schools for STEM programs, or increasing teacher knowledge of and exposure to STEM. To my knowledge, the strategies I put forth are the first to suggest an intersection between the DoD, STEM companies, and universities. The two strategies I proposed will not only help mitigate the shortage of domestic STEM students, but will benefit STEM companies, universities, domestic students, and the Department of Defense alike.

Other Ways in Which the Topic of Increasing Domestic Student STEM Participation can be Explored

While I took a very specific approach to addressing this issue, there are many opportunities for exploring other solutions and strategies, and for conducting further research. One possible solution that has been discussed in current literature, but should be further explored, is to provide more H-1B visas for foreign STEM workers. Currently, the DoD issues a limited number each year, only 65,000 to 85,000 (*H-1B Specialty Occupations*). Granted, there are likely valid reasons for the visa cap, but it would be worth exploring whether the DoD could allocate a higher number of H-1Bs every year. This would allow the United States to bring in more foreign STEM talent capable of working on defense projects, helping to fill the current gap between the DoD's needs and the supply of qualified domestic or H-1B-holding workers.

Another possible strategy that could be further explored is universities partnering to offer joint admissions programs for STEM FSDPs. If a smaller university or college is unable to offer

an FSDP, that university could partner with a larger university that does offer FSDPs. This way, students could seamlessly transition from one university to another in order to obtain their FSDP. Certain universities have already established similar joint admissions programs that have been successful, such as Carlow University did with UPMC's School of Nursing (SON). In this case, students who enroll in UPMC SONS's RN⁸ Diploma Program are assured of admission and a seamless transition into Carlow's RN to BSN⁹ program (*Joint Admissions Agreement, UPMC Schools of Nursing*). If universities could do something similar for STEM FSDPs, this could possibly help attract and retain domestic STEM students.

I believe it would also be beneficial to further explore how to increase high school students' exposure to STEM, as well as improving the quality of current exposure. As of now, current literature on this topic is existent, though limited. If the United States could develop methods to increase students' exposure to STEM and its related fields while they are still in high school, this could help increase domestic students' interest in pursuing a STEM degree in college. Moreover, research has shown that the current quality of high school STEM education is poor and severely lacking. Finding ways to improve the quality of our high school STEM education could also help increase students' exposure to and comfortability with STEM while in high school, so that they feel more inclined and confident to pursue STEM in college and beyond.

⁸ Registered Nurse

⁹ Bachelor's of Science in Nursing

Bibliography

Asp, David. "Espionage Act of 1917 (1917)." The Free Speech Center, 6 Aug. 2024, www.firstamendment.mtsu.edu/article/espionage-act-of-1917/#:~:text=The%20Espionage%20Act%20of%201917%20prohibited%20obtaining%20information%2C%20recording%20pictures,advantage%20of%20any%20foreign%20nation. Accessed 24 Feb. 2025.

Boggs, George R., et al. "Addressing the Stem Workforce Shortage." U.S. Chamber of Commerce Foundation, U.S. Chamber of Commerce, 25 June 2024, www.uschamberfoundation.org/education/addressing-stem-workforce-shortage. Accessed 17 Nov. 2024.

This report was published in late 2022 by the U.S. Chamber of Commerce Foundation, the research and education arm of the U.S. Chamber of Commerce. It was written in response to the current labor shortage in the STEM fields. It also provides insight as to some current attempts to reduce this shortage, namely funding for community colleges to support STEM education. Specifically, the National Science Foundation (NSF) provides \$1.11 billion in support to community colleges through their Advanced Technology Education program (ATE). This report is relevant to my thesis as it provides key data regarding the shortage of STEM workers.

Boğar, Yurdağül, and Jari Lavonen. "Status and Trends of STEM Education in Finland." *Research Gate*, Oct. 2022, www.europarl.europa.eu/pdf/divers/week-26-2016/budg-30-06-2016.pdf. Accessed 10 Dec. 2024.

Conducted by the Finnish government, this research study examines the methods Finland other STEM-strong countries have implemented and continue to implement to foster participation and proficiency in the STEM fields. The impetus for this study is to provide other countries, particularly those falling behind in STEM, with potential strategies for increasing citizen participation. The goal of this paper is like that of my thesis, except that it is not targeted towards providing strategies for one specific country. Whereas my thesis specifically focuses on increasing citizen participation in the United States. This study provides part of the foundation for the two strategies that I propose in my thesis.

Burke, Abigail Okrent and Amy. "Science & Engineering Indicators." NSF, <https://nces.nsf.gov/pubs/nsb20212>. Accessed 17 Nov. 2024.

This 2024 report published by National Science Foundation (NSF) details the current state of STEM education and labor force in the United States, especially within a global context. The summary report is constituted of nine thematic reports on all-degree levels of STEM education. The thematic reports also cover the STEM work force and public perceptions and awareness towards STEM, among other topics. The data from the thematic reports is categorized into three groups: talent, discovery, and translation. These groups are interconnected, in that talent in the fields of STEM prompts scientific discovery, which is in turn translated to our society and economy through innovation. This report provides relevant data to my paper, specifically regarding the shortage of qualified civilian STEM workers. For instance, data from the report shows that an overwhelming majority of STEM scientists and engineers in the U.S. are foreign-born individuals that possess neither a citizenship nor an H-1B Visa.

“23 Careers You Can Get with a Computer Science Degree.” Central Michigan University, 1 Dec. 2023, www.cmich.edu/blog/all-things-higher-ed/23-careers-computer-science-degree. Accessed 8 March 2025.

Cohen, Carolyn, et al. “Fostering Stem Career Awareness.” *Northwest Association for Biomedical Research*, [nwabr.org/sites/default/files/pagefiles/PDK STEM Career Awareness.pdf](http://nwabr.org/sites/default/files/pagefiles/PDK%20STEM%20Career%20Awareness.pdf). Accessed 17 Mar. 2025.

Daker, Richard, et al. “(PDF) First-Year Students’ Math Anxiety Predicts Stem Avoidance and Underperformance throughout University, Independently of Math Ability.” *Research Gate*, Nature Partner Journals: Science of Learning, 14 June 2021, [www.researchgate.net/publication/352378451_First-year students’ math anxiety predicts STEM avoidance and underperformance througout university independently of math ability](http://www.researchgate.net/publication/352378451_First-year_students'_math_anxiety_predicts_STEM_avoidance_and_underperformance_througout_university_independently_of_math_ability). Accessed 8 Dec. 2024

This study, published in June of 2021, researched the link between the phenomenon of math anxiety and students’ attitudes towards and performance in STEM. Prior research on the links between math anxiety and students’ STEM outcomes (i.e. likelihood of pursuing STEM fields and their performance in those fields) was very limited. Daker and his peers’ study significantly contributed to the literature on this topic. Their study proved that math anxiety influences STEM outcomes independent of students’ math ability. This means that, even if students understand mathematical concepts, their developed anxiety towards math will still influence them to avoid pursuing STEM fields. The contribution of this study to my thesis is that it links math anxiety as one of the causes of the shortage of STEM students in universities.

Deelen, Grace van. “National Science Board Reports a Need for More Support of STEM Talent.” *Eos*, 26 Aug. 2024, www.eos.org/articles/national-science-board-reports-a-need-for-more-support-of-stem-talent#:~:text=The%20report%20emphasized%20that%20flat,scientists%20and%20engineers%20from%20abroad. Accessed 24 Feb. 2025.

Dwyer, Morgan, et al. To Compete, Invest in People: Retaining the U.S. Defense Enterprise’s Technical Workforce. Center for Strategic and International Studies (CSIS), 2020. JSTOR, <http://www.jstor.org/stable/resrep27055>. Accessed 1 Dec. 2024.

In 2020, the Center for Strategic and International Studies (CSIS), a nonprofit research organization focused on developing practical solutions to addressing major global issues, published this article in response to the staffing struggles experienced by the U.S. defense industry. Back in 2020, and even still today, the U.S. defense industry is struggling to hire the technical talent needed to keep up with the rapid modernization and development of defense technology. The article puts forth several strategies to reduce the industry’s hiring difficulties and the resultant shortage of employees. These strategies include but are not limited to: including STEM professionals in the hiring process; clearly defining requirements for future STEM talent; and further training the existing workforce to develop new technical skills. This article provides me important data and recommendations for my section on the impacts of student underperformance and lack of participation in STEM education on the fields of STEM, and how to best mitigate these impacts.

“Employee Benefits.” *Go Government*, 5 Dec. 2024, <https://gogovernment.org/all-about-government-jobs/employee-benefits/#:~:text=Through%20the%20Public%20Service%20Loan,or%20another%20a%20public%20service>. Accessed 17 January 2025.

This website provides effectively any information one would need regarding government jobs. The website was created by a non-partisan, non-profit organization, Partnership for Public Service. The purpose of this organization is to provide resources to ease the process of finding and onboarding a government job. Partnership for Public Service works closely with the federal government, so the information on the website is reliable and accurate. For the purposes of my thesis, I only utilized the section entitled “Employee Benefits.” This was the source of the information I included in my fourth section regarding the benefits of government jobs. The “Employee Benefits” section provides a very detailed overview of the myriad benefits government workers receive.

Falloon, Garry, et al. “Building STEM in Schools: An Australian Cross-Case Analysis.” *Educational Technology & Society*, vol. 24, no. 4, 2021, pp. 110–22. JSTOR, <https://www.jstor.org/stable/48629249>. Accessed 30 Nov. 2024.

This article, published by the quarterly academic journal, *Educational Technology & Society*, analyzes data from The Principals as STEM Leaders (PASL) project. The PASL, initially launched in 2017, was a national research and professional learning program for school principals funded by the Australian Government. The goal of the PASL was to develop more expansive and robust STEM leadership in the Australian education system. This article also examines the approaches four different Australian schools took towards improving its students’ performance in STEM education. The data and findings provided by this article offer valuable insights as to the various ways I can synthesize already established and effective approaches to STEM education into a practical approach that the U.S. can implement to improve student performance in STEM education.

“Former Air National Guardsman Sentenced to 15 Years in Prison for Unlawfully Disclosing Classified National Defense Information.” Office of Public Affairs | Former Air National Guardsman Sentenced to 15 Years in Prison for Unlawfully Disclosing Classified National Defense Information | United States Department of Justice, 6 Feb. 2025, www.justice.gov/archives/opa/pr/former-air-national-guardsman-sentenced-15-years-prison-unlawfully-disclosing-classified. Accessed 24 Feb. 2025.

Francis, Reuben. “Most International Graduates of American Universities Ultimately Leave the U.S.” Economic Innovation Group, 2025, www.eig.org/immigrant-retention-estimates/. Accessed 17 March 2025.

Graham, Mark J, et al. “Science Education. Increasing Persistence of College Students in STEM.” *Science (New York, N.Y.)*, U.S. National Library of Medicine, 9 May 2023, <https://pmc.ncbi.nlm.nih.gov/articles/PMC10167736/#R5>. Accessed 20 Jan. 2025

This research study, published by the National Library of Medicine in May of 2023, provides a model for increasing student persistence in STEM majors. Many universities are experiencing high attrition rates in STEM majors. In response to this phenomenon, Graham et al.’s study examines various prior studies to develop their own model, referred to in the paper as a “persistence framework,” to help universities reduce high STEM major attrition rates. The study also briefly examines several causes for STEM student attrition. It also analyzes successful

retention programs and their methods. My thesis utilizes data from this study to show the severity of the shortage of domestic STEM students.

Grays, Kameron. “The STEM Landscape in 2024: Key Trends and Statistics.” *STEM DTS*, 25 July 2024, <https://stemdts.com/the-stem-landscape-in-2024-key-trends-and-statistics/#:~:text=STEM%20careers%20continue%20to%20offer,expertise%20required%20in%20these%20areas>. Accessed 18 Jan. 2025.

This article, published by STEM Diverse Talent Solutions (DTS) in 2024, a company dedicated to connecting prospective employees with leading STEM companies, provides a comprehensive list of key statistics regarding trends in STEM in 2024. The article examines these trends to provide insight into the current state of STEM, as well as predict what the future of STEM in the United States may look like. Data from this article show that median annual wages for STEM jobs are significantly higher than those of non-STEM occupations. My thesis uses this data to show one of the numerous incentives for pursuing a STEM career.

Hill, Heather C., et al. “Professional Development That Improves STEM Outcomes.” *The Phi Delta Kappan*, vol. 101, no. 5, 2020, pp. 50–56. JSTOR, <https://www.jstor.org/stable/26898130>. Accessed 29 Nov. 2024.

Written by researchers at Phi Delta Kappa International, a non-profit organization dedicated to supporting teachers and school leaders, this paper conducted a meta-analysis of research on STEM teacher development programs- referred to as professional development (PD) programs in the paper. Essentially, PD programs seek to better a teacher’s understanding of classroom content and curricula, instructional materials, and state standards, among other topics. The paper found that many STEM teachers, despite participating in PD programs, lack what the authors refer to as the “professional knowledge” needed to help students and improve their academic outcomes. The authors define this professional knowledge as encompassing an instructor’s understanding of how the content they are teaching, how the students think about and absorb this content, and curriculum itself, all tie together. This professional knowledge also includes the teacher’s ability to make good instructional decisions based on specific situations. The paper concluded that PD programs that better teachers’ professional knowledge are correlated with higher student performance. This paper supports the idea reflected in my thesis that one of the avenues to bettering student performance in STEM education is through improving the quality of the teachers.

Howell, Sam. *Technology Competition: A Battle for Brains*. Center for a New American Security, 2023. JSTOR, <http://www.jstor.org/stable/resrep52276>. Accessed 30 Nov. 2024.

Published by the Center for New American Security (CNAS), a nonprofit organization focused on developing pragmatic and ethical policies for national security and defense, this paper examines whether the U.S. has the STEM talent to compete with its global competitors when it comes to emerging technologies. The paper also evaluates if the current methods of developing STEM talent are sufficient to meet the demands of the STEM workforce. The author concludes by laying out several recommendations as to how improve U.S. STEM education and increase student interest in pursuing jobs in the fields of STEM. Among the many recommendations Howell sets forth, one that I will likely discuss in my paper is her suggestion that universities partner with independent organizations to develop industry-specific educational programs. This

paper is important to my thesis, as it provides key examples of the effects of low student participation and performance in STEM education on the fields of STEM.

“H-1B Specialty Occupations.” *USCIS*, 17 Jan. 2025, www.uscis.gov/working-in-the-united-states/h-1b-specialty-occupations#:~:text=The%20H%2D1B%20classification%20has,H%2D1B%20Cap%20Season%20page.&text=When%20can%20I%20begin%20working,period%20of%20authorized%20stay%20expires. Accessed 21 Jan. 2025.

This section on the U.S. Citizenship and Immigration Services’ website provides extensive information on the H-1B Visa program, how to apply for an H-1B Visa, and so on. It also provides details on the yearly H-1B cap. Each year, the Department of Defense restricts how many H-1B Visas they grant to foreign workers. The cap is 65,000, with an additional 20,000

available for obtainment under specific circumstances, for a total of 85,000 available H-1B Visas a year. My thesis drew data specifically from this section, per my discussion of increasing the annual H-1B cap in my section on recommendations for further research.

“The H-1B Visa Program and Its Impact on the U.S. Economy.” *American Immigration Council*, 2025, <https://www.americanimmigrationcouncil.org/research/h1b-visa-program-fact-sheet>. Accessed 22 Jan. 2025.

Published by the American Immigration Council in January of 2025, this article, similar to the website discussed above, lays out what an H-1B Visa is, how to petition for one, and the impact of the H-1B Visa Program on the United States. It also provides data on the annual H-1B Visa cap, as well as the denial rate of petitions for H-1B Visas per fiscal year. While the information in the article is extensive, my thesis specifically utilized the article’s definition of an H-1B Visa to better explain to readers the specifics of the purpose of an H-1B Visa, who can apply for one, and the circumstances in which it is useful.

Insler, Michael, et al. *Fear and Loathing in the Classroom: Why Does Teacher Quality Matter?* IZA - Institute of Labor Economics, 2021. JSTOR, <http://www.jstor.org/stable/resrep61578>. Accessed 29 Nov. 2024.

This research paper, written in 2021 by several researchers at the U.S. Naval Academy, Lehigh University, and the IZA Institute of Labor Economics, examines the correlative relationship between teacher quality and student learning and performance. The data from the study shows that the quality of the teacher has a direct, profound impact on the student’s academic performance. To test this, the study analyzed two types of teachers. The first type are teachers who help boost students’ common final exam scores, which are objectively graded by a committee, as opposed to the teacher themselves. The second are teachers who give easier subjective grades on their students’ exams. The former were proven to enhance student performance, while the latter to impair it. This paper supports the claim reflected in my thesis that one of the strategies for improving student performance in STEM education involves improving teacher quality and holding teachers to higher standards.

Joint Admissions Agreement, UPMC Schools of Nursing, 17 Mar. 2015, www.carlow.edu/wp-content/uploads/2021/10/upmc-schools-of-nursing-rn-bsn.pdf. Accessed 21 Jan. 2025.

Published on Carlow University’s website in 2015, this document details the joint admissions agreement between the University of Pittsburgh Schools of Nursing and Carlow University for

students in the Registered Nurse (RN) Diploma programs at UPMC's Schools of Nursing to enter directly into Carlow's Registered Nurse into Bachelor of Science in Nursing (RN-BSN) degree program. I used this agreement between the two universities as an example of potentially beneficial types of partnerships that should be considered for further research on increasing domestic student participation in STEM. The purpose of this agreement between UPMC and Carlow was to increase participation in the RN-BSN programs, and to service UPMC hospitals with a greater number of highly qualified nurses. This suggests that universities could establish similar partnerships for STEM programs, and that this could potentially increase student participation in STEM.

Laughlin, Lynda, et al. "Technician Is a Common Job Among Young STEM Workers." *United States Census Bureau*, Nov. 2022,

www.census.gov/library/stories/2022/11/stem-workers-under-age-25.html#:~:text=Aging%20STEM%20Workforce%20Still%20Younger,the%20next%20generation%20of%20workers. Accessed 12 March 2025.

This report was published in early 2024 by the Boston Consulting Group (BCG), a firm that partners with businesses to address the pressing challenges they face. Written in conjunction with SAE International, a worldwide organization of over 128,000 engineers and other such related technical experts, the report addresses the current shortage of well-educated, citizenship- or H-1B Visa- holding engineers in the United States. Analyzing data from the Bureau of Labor Statistics (BLS), the report explains the risk this shortage poses to the U.S. economy and GDP. It also elaborates on challenges that hinder the U.S. from properly addressing this shortage and possible methods of overcoming these challenges. The pertinence of this report to my thesis is it provides much of the backbone for the second section of my paper on the implications of student underperformance in STEM education and the shortage of qualified STEM workers.

Laughlin, Lynda, et al. "Technician Is a Common Job Among Young STEM Workers." *United States Census Bureau*, Nov. 2022, [www.census.gov/library/stories/2022/11/stem-workers-under-age-](https://www.census.gov/library/stories/2022/11/stem-workers-under-age-25.html#:~:text=Aging%20STEM%20Workforce%20Still%20Younger,the%20next%20generation%20of%20workers)

[25.html#:~:text=Aging%20STEM%20Workforce%20Still%20Younger,the%20next%20generation%20of%20workers](https://www.census.gov/library/stories/2022/11/stem-workers-under-age-25.html#:~:text=Aging%20STEM%20Workforce%20Still%20Younger,the%20next%20generation%20of%20workers). Accessed 12 March 2025.

"Long-term." Merriam-Webster.com Dictionary, Merriam-Webster, <https://www.merriam-webster.com/dictionary/long-term>. Accessed 10 Mar. 2025.

Lopez, C. Todd. "Defense Department Engineers Drive Mission Success." U.S. Department of Defense, www.defense.gov/News/News-Stories/Article/Article/2942301/defense-department-engineers-drive-mission-success/#. Accessed 27 Nov. 2024.

In 2022, the United States Department of Defense published this article regarding the need for qualified, citizenship-holding STEM workers. It also explains the impacts of the current shortage of such employees on the fields of STEM, specifically in regard to U.S. government defense projects. The article details current efforts by the Department of Defense to support universities, engineering, and STEM education, such as certain scholarship programs and fellowships. This article contributes to informing the second section of my paper, which explains the impacts the shortage of qualified native-born engineers on the fields of STEM, and, therefore, on our country and economy as whole.

Lytle, Ashley, and Jiyun Elizabeth Shin. "Incremental Beliefs, STEM Efficacy and STEM Interest Among First-Year Undergraduate Students." *Journal of Science Education and Technology*, vol. 29, no. 2, 2020, pp. 272–82. JSTOR, <https://www.jstor.org/stable/48741997>. Accessed 30 Nov. 2024.

This paper, published in the *Journal of Science Education and Technology*, examines the low retention rates of undergraduate and graduate students pursuing degrees in STEM, and the problems this poses for the United States. The reason low undergraduate and graduate retention in STEM is an issue is that participation and retention of students in the fields of STEM is vital to ensuring that STEM jobs are adequately staffed. Currently, the STEM industry is expanding, and an increasing number of job opportunities are being created. Yet too few students are graduating with degrees in STEM. Thus, the STEM fields are lacking qualified employees to fill the increasing number of STEM positions in the workforce. The paper provides several recommendations for future directions and implications of this concerning trend among collegiate-level STEM students. This paper supports my argument that the U.S. is lacking qualified civilians in the STEM fields, and that this poses several problems for our country and economy.

Malcom, Shirley. *Barriers and Opportunities for 2-Year and 4-Year STEM Degrees: Systemic Change to Support Students' Diverse Pathways.*, U.S. National Library of Medicine, 18 May 2016, www.ncbi.nlm.nih.gov/books/NBK368186/#sec_000076. Accessed 24 Feb. 2025.

Marginson, Simon, et al. (PDF) *Stem: Country Comparisons: International Comparisons of Science, Technology, Engineering and Mathematics (STEM) Education. Final Report.*, www.researchgate.net/publication/305063165_STEM_country_comparisons_international_comparisons_of_science_technology_engineering_and_mathematics_STEM_education_Final_report. Accessed 17 Nov. 2024.

Conducted by the Australian Council of Learned Academies (ACOLA), this research study examines the data from 22 specific reports they commissioned worldwide. It aims to see what methods other countries are implementing to foster participation and proficiency in the fields of STEM. The impetus for this study is to find strategies to improve STEM participation and proficiency in Australia, as the country is falling behind other developed nations. The goal of this paper is similar to that of my thesis, except my thesis is focused on the U.S. instead of Australia. This study provides much of the data that I use in my thesis, particularly in regard to the methods and strategies other first-world countries are implementing to foster strong participation and performance in STEM.

"National Foundation for American Policy - NFAP Policy Brief." *National Foundation for American Policy*, July 2013, <https://www.nfap.com/pdf/New%20NFAP%20Policy%20Brief%20The%20Importance%20of%20International%20Students%20to%20America,%20July%202013.pdf>. Accessed 7 Dec. 2024.

This report from the National Foundation for American Policy (NFAP) provides extensive data on how heavily the U.S. STEM industry relies on foreign STEM talent. As discussed in my thesis, international students currently comprise the majority of graduate students in STEM, as well as the majority of the workforce. My thesis uses data from this report to support this fact. Specifically, I use the example of engineering programs, which are heavily dominated by foreign

students. Similar to the recommendation section of my thesis, this report also calls for an increase in the number of H-1B Visas allocated to foreign STEM talent each year.

Neufeld, Jeremy. “STEM Immigration Is Critical to American National Security.” *Institute for Progress*, Mar. 2022, www.ifp.org/stem-immigration-is-critical-to-american-national-security/. Accessed 10 March 2025.

Olis, Brendan, et al. “The Global Distribution of STEM Graduates: Which Countries Lead the Way?” Center for Security and Emerging Technology, 29 Mar. 2024, <https://cset.georgetown.edu/article/the-global-distribution-of-stem-graduates-which-countries-lead-the-way/#:~:text=Moreover%2C%20just%2020%20percent%20of,behind%20both%20Mexico%20and%20France>. Accessed 8 Dec. 2024.

Published in March of 2024 by the Center for Security and Emerging Technology (CSET), a policy research organization affiliated with Georgetown University’s Walsh School of Foreign Service, this article explores which countries are currently dominating on the global STEM stage. Specifically, the article examines which countries are graduating the most STEM students, relative to the overall number of graduates across all fields. My thesis uses data from this article to show that China is graduating far more STEM students, percentagewise, than the United States. As explained in my thesis, this is problematic for the United States, as China is advancing in the STEM fields far faster than the U.S., particularly with the recent development of DeepSeek AI, which singlehandedly overthrew the U.S. as the world’s leader in AI overnight.

Rosenzweig, Emily Q., and Xiao-Yin Chen. “Which Stem Careers Are Most Appealing? Examining High School Students’ Preferences and Motivational Beliefs for Different Stem Career Choices - International Journal of STEM Education.” *SpringerOpen*, Springer International Publishing, 9 June 2023, <https://stemeducationjournal.springeropen.com/articles/10.1186/s40594-023-00427-6#citeas>. Accessed 19 Jan. 2025.

This study examines high school students’ perceptions of STEM and their likelihood of pursuing certain STEM degrees and careers, as well as what influenced their preferences. As a subset of this research, the study also sought to determine if discrepancies in preferences existed between students of different races, ethnicities, and genders. The study showed that students were almost entirely unaware of any STEM fields beyond those that are generally well-known, and that they were less likely to pursue a degree in a field they perceived themselves as unlikely to succeed in. Further, this study proved that students’ gender and racial/ethnic identities impacted their preferences and opinions towards certain STEM fields versus others. My thesis uses data from this study to show that students belonging to racial minorities, as well as female students, are less likely to pursue STEM degrees and careers, due to underrepresentation and concerns of discrimination.

“Security Clearances for Non-U.S. Citizens.” *Clearance Jobs*, www.support.clearancejobs.com/security-clearance-faqs/security-clearances-for-non-u-s-citizens. Accessed 17 Mar. 2025.

Sentient Digital, Inc. “The Benefits of University Partnerships with Companies: Key Takeaways from Sentient Digital and RDA at Villanova.” *Sentient Digital, Inc.*, 11 May 2024,

<https://sdi.ai/blog/the-benefits-of-university-partnerships-with-companies-key-takeaways-from-sentient-digital-and-rda-at-villanova/>. Accessed 10 Jan. 2025.

This section on Sentient Digital's website, a company dedicated to furthering innovation in the defense and aerospace sectors of STEM, details a connection made between Sentient Digital and Villanova University. Representatives from Sentient Digital's subsidiary, RDA, attended various career fairs at Villanova, for the purpose of establishing a strong connection between RDA and Villanova. Furthermore, this section explains the numerous benefits of companies partnering with universities. Perhaps the most notable of these benefits are the funding companies can provide universities for the purpose of research, as well as the direct source of qualified employees that companies are able to establish. My thesis uses information from this section of Sentient Digital's website to support my claim that both universities and companies could greatly benefit from establishing closer partnerships.

Skrentny, John David. *Wasted Education: How We Fail Our Graduates in Science, Technology, Engineering, and Math*. The University of Chicago Press, 2023. Accessed 25 Nov. 2024.

In his book, *Wasted Education*, Dr. John D. Skrentny, a Harvard graduate with a doctorate degree in sociology, addresses the problems with STEM education in the United States, and the impact it has on the country and economy. Dr. Skrentny discusses at length the lack of students pursuing STEM degrees in higher education, at both undergraduate and graduate levels, as well the implications of this phenomenon. The lack of students pursuing STEM degrees is one of the direct causes of the shortage of qualified civilian engineers. Dr. Skrentny also elaborates on another current phenomenon, which he terms "The Exodus from STEM Jobs." The phenomenon Skrentny is describing is the fact that an overwhelming majority of STEM graduates are entering non-STEM occupations, which typically offer higher pay than those in STEM. This book provides the statistical backbone of much of my thesis, especially regarding participation in STEM and some of the specific ways in which the U.S. STEM education system is lacking.

Smith, Simone. "Stem Career Awareness for Younger Students Enables Academic and Workforce Success." *STEM Career Awareness for Younger Students Enables Academic and Workforce Success - Florida Association of School Administrators*, 13 Jan. 2025, www.fasa.net/Leader2Leader/stem-career-awareness-for-younger-students-enables-academic-and-workforce-success-. Accessed 17 March 2025.

Stewart, Jennifer. "Vital Signs 2024: U.S. Defense Industrial Base Still Facing Headwinds." *National Defense Magazine*, 5 Apr. 2024, www.nationaldefensemagazine.org/articles/2024/4/5/vital-signs-2024-us-defense-industrial-base-still-facing-headwinds. Accessed 17 March 2025.

Taylor, Danielle, and Caren A. Arbeit. "Science & Engineering Indicators." *NSF*, May 2024, <https://nces.nsf.gov/pubs/nsb20245/u-s-stem-workforce-size-growth-and-employment>. Accessed 3 Jan. 2025.

This report provides key data and statistics regarding the state of the United States STEM workforce. The report covers a wide variety of topics relating to the STEM workforce, from size, growth, and employment to representation of demographic groups. My thesis utilized much of the data and statistics provided by this report. In the first section of my thesis, I use data from this report to help show how severely women and racial minorities are underrepresented in the STEM workforce. Then I use this report's data to show that the majority of STEM workers in the

United States do not possess a bachelor's degree or higher. This is just one of the ways in which the shortage of students pursuing STEM degrees is reflected in the workforce.

“Top Secret / Sensitive Compartmented Information (TS/SCI) Clearance.” *Technology Transformation Services Handbook*, <https://handbook.tts.gsa.gov/general-information-and-resources/business-and-ops-policies/top-secret/>. Accessed 18 Jan. 2025.

This section in the U.S. Government's *Technology Transformation Services Handbook* details the process and timeline of receiving one's government security clearances. Depending on the level of clearance, the timeframe will be anywhere from one month for the lowest level to as long as almost a year and a half for the highest level. In the fourth section of my thesis, I use the information from this website to support my argument as to why the Department of Defense should establish Domestic Student Pre-Screening (DSPS) as part of Field Specific Degree Programs. This way, STEM students can begin the process of obtaining their DoD security clearances before even entering the workforce.

Turner, John. “Report: Pentagon Says STEM Education Deficit Is Weakening America.” *The National Interest*, 7 Feb. 2021, www.nationalinterest.org/blog/buzz/report-pentagon-says-stem-education-deficit-weakening-america-177680. Accessed 5 March 2025.

von Schirach, Paolo. “China Beats America in STEM Education.” *China Beats America in STEM Education* | GPI, Global Policy Institute, Sept. 2024, <https://globalpi.org/research/china-beats-america-in-stem-education/#:~:text=China%20is%20massively%20ahead%20of%20the%20U.S.,STEM%20include%20Science%2C%20Technology%2C%20Engineering%20and%20Mathematics>. Accessed 20 Jan. 2024.

This brief explains the urgency of mitigating the shortage of domestic STEM students in the United States, as China is far surpassing the U.S. in this regard. Worldwide, China graduates the highest percentage of STEM students from their universities, which are often outfitted with high-tech laboratories and research facilities. The brief calls for the United States to respond in a similar manner to how they responded during the Cold War when the USSR launched Sputnik. My thesis uses this article to help show that mitigating the lack of domestic STEM students should be the U.S. government's highest priority right now.

“Why Don't Students Stick with STEM Degrees?” *Stevens Institute of Technology*, 2023, www.stevens.edu/news/why-dont-students-stick-with-stem-degrees. Accessed 20 Jan. 2025.

Published by the Stevens Institute of Technology in 2023, this article explains why domestic students in the United States fail to complete STEM degrees in their chosen field. The article also shows that students belonging to racial minorities are far less likely to pursue and graduate with STEM degrees than white students. Further, it explores various methods universities can implement to attract and retain domestic students. This article helps support my explanation of why domestic students are unlikely to pursue and complete STEM degrees, as well as to show the specific attrition and retention rates on average across American universities.

WRT - 1077 toward Collective Impact in DOD's STEM Activities and Outreach, https://dodstem-assets.dodstem.us/files/DoD_STEM_Collective_Impact_Report_SERC_2024.pdf#page155. Accessed 18 Nov. 2024.

In January of 2024, the United States Department of Defense, in conjunction with the Stevens Institute of Technology, published a two-year long research study on where U.S. STEM education is lacking and what the DoD can do to help. The study examines what strategies other developed nations are using to boost performance and participation in STEM. It also cites several case studies conducted throughout the United States, examining state-wide initiatives to combat student underperformance on standardized mathematics and science tests and the labor shortage in the fields of STEM. This extensive 177-page research study provides me with an abundance of data for my paper, especially regarding the methods implemented by other countries to develop and maintain their robust STEM education systems and high levels of performance and participation in STEM fields.